

Team Name: SixTeam

By: Germin, Sean, Angel, Aidan

System Components & Breakdown

1. Frontend (User Interface)

- **Components:** Meal Planning Dashboard, Recipe & Video Section, Input Forms, Barcode/Photo Scanner Interface.
- **Tools/Frameworks:**
 - **Language:** HTML, CSS
 - **Framework:** React (fast, component-based, easy state management, integrates well with API)

2. Backend (Application Layer)

- **Components:** Authentication & User Profiles, Meal Planning Feature, Nutrition & Budget Calculator, Recipe Recommendation System, Integration APIs.
- **Tools/Frameworks:** **Language:** Python/SQL, **Framework:** Django

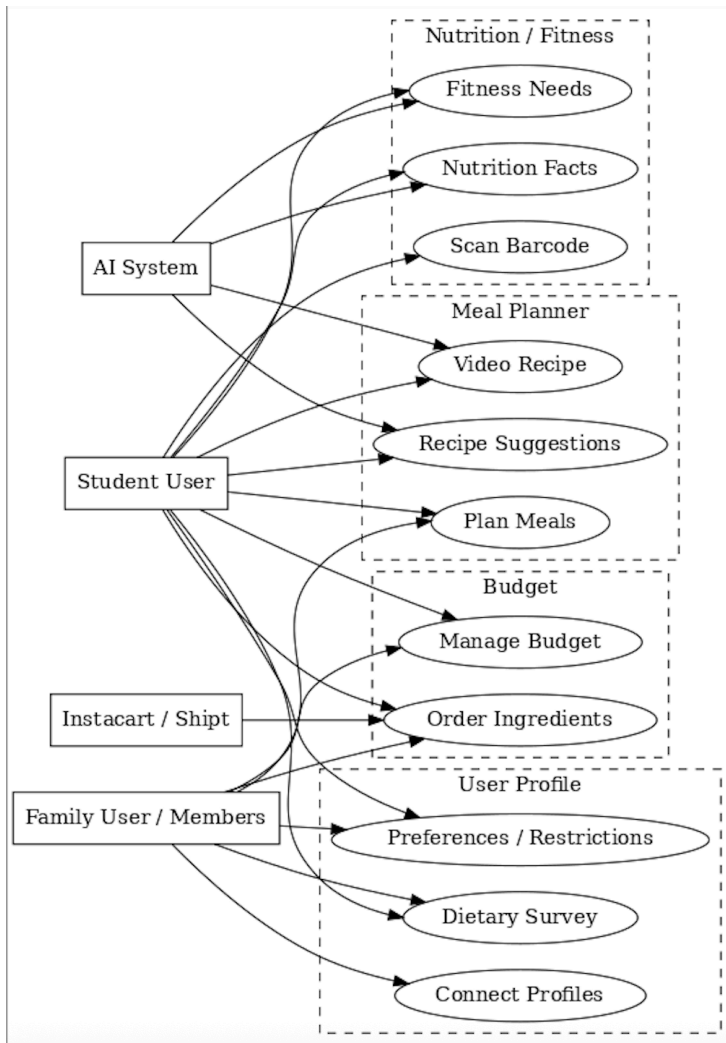
3. Database

- **Components:**
 - User Data:** profiles, preferences, restrictions, saved plans.
 - Meal & Recipe Data:** ingredients, nutrition info, video links.
 - Budget Data:** cost per serving, pricing from stores.
 - Scanned Data:** barcode lookup history, uploaded food images (metadata stored in DB).
- **Tools:** SQL Database, Django

4. External Integrations

- **Nutrition APIs**
- **Barcode Scanning APIs**
- **Video Hosting APIs**
- **Fitness Apps (Optional)**

Use Case Diagram



Requirements:

Nutrition/Fitness

- **Fitness Needs:**
 - The user must be able to see what the recommended amount of calories, macronutrients, micros, etc., is after selecting their preferences and inputting their weight, height, and roughly how much exercise.
- **Nutrition Facts:**
 - The user, upon clicking on an item, must be able to view all the nutrition facts, mainly showing: calorie count, protein, carbs, and fat.
- **Scan Barcode:**
 - The user should be able to use their phone and scan the barcode to identify what item it is, and display the price (if linked to any grocery store), along with the nutrition facts from that one item

Nonfunctional Requirements

- **Performance**
 - The system must be able to run smoothly and efficiently
- **Reliability**
 - System must be able to recommend (with use of AI) recipes that suit the user
- **Availability**
 - System must be able to give what the User needs, in this case recipes that suit their preferences.

Meal Planner

- ***Video Recipes:***
 - The user should provide the system with a prompt that provides details on the desired recipe.
 - The system must provide the user with at least 1 or more link to video recipes that fit the description that the user requested.
 - The system should provide at least 1 or more link to video recipes that are based on the user's preferences described in the user's profile.
 - The system should use an LLM to provide the suggestion.
- ***Recipe Suggestions:***
 - The user should provide the system with a prompt that provides details on the desired recipe.
 - The system must provide the user with at least 1 or more recipe options that adhere to the user's request.
 - The system must display the full recipe to the user.
 - The system should additionally provide an external link to a recipe if the recipe was not generated by an LLM.
- ***Plan Meals:***
 - The system must provide data fields where the user can input different types of meals that they intend to eat daily.
 - The system must display the user's meal plan in a table.

- The system should have the option to display the user's meal plan by day, week, month, quarter, and year.

Meal Planner Non-Functional Requirements:

- **Usability:**
 - The system must be run using a stable internet connection.
 - The system must be connected to the LLM and be able to receive and send information.
- **Security:**
 - The system database access must be limited to strictly necessary features.
 - The system database must be secure from any unintended access or attacks.
- **Reliability:**
 - User information must be saved to a reliable database and accessible to the application.

User Profile Functional Requirements

- ***Preferences/Restrictions:***
 - The user should disclose dietary preferences and restrictions
 - The system must store this data in a profile
 - The system must be able to create multiple different profiles with different dietary preferences
 - The system should recommend meals based on preferences in profiles
 - The system must recommend meals based on restrictions in profiles
- ***Dietary Survey***
 - The system must provide a survey for the user inquiring about dietary restrictions and preferences
 - The user should fill in the survey to optimize system functionality
- ***Connect Profiles***
 - The user should be able to create multiple profiles
 - The user should be able to select which profiles the system should use when making recommendations
 - The system should be able to detect all profiles or a combination of profiles when making recommendations

User Profile Non-Functional Requirements

- **Usability:**
 - The system must be compatible with any current web browser, users should be able to access their profiles in under 3 seconds
- **Security:**
 - The system should keep user data secure to only the app, providing essential data (credit card information) only for necessary instacart/shipt purchases
 - The system should be transparent about how user data is used
- **Reliability**
 - The system should run well using any current web browser

Budget Calculator

The Budget Calculator feature helps users estimate and manage grocery expenses when shopping online through Instacart or Shipt.

1. Functional Requirements

1.1 Budget Input & Management

- Users shall be able to **set a total budget** for groceries.
- Users shall be able to **update or reset** their budget at any time.

1.2 Shopping Cart Integration

- The system shall integrate with **Instacart or Shipt APIs** to fetch real-time cart data (item prices, fees, discounts, and taxes).
- The system shall **calculate the total** of the current shopping cart against the user's set budget.

1.3 Budget Tracking & Alerts

- The system shall display a **remaining balance** after each item is added to the cart.
- The system shall provide **real-time alerts** if the cart total exceeds the budget.

1.4 Reporting & Insights

- Users shall be able to **view budget history** to analyze spending over time.

1.5 User Accounts & Profiles

- Users shall be able to **log in** with a secure account.
- Users can **connect multiple family members' profiles**, with shared or individual budgets.

2. Non-Functional Requirements

- **Performance:** Budget calculations and cart updates shall reflect changes within 2 seconds.
- **Scalability:** The system shall support integration with multiple vendors beyond Instacart/Shipt in the future.

- **Security:** API connections shall use secure authentication.
- **Usability:** The interface shall be mobile-friendly and accessible.